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14.1.1: Complex Number Addition using Class

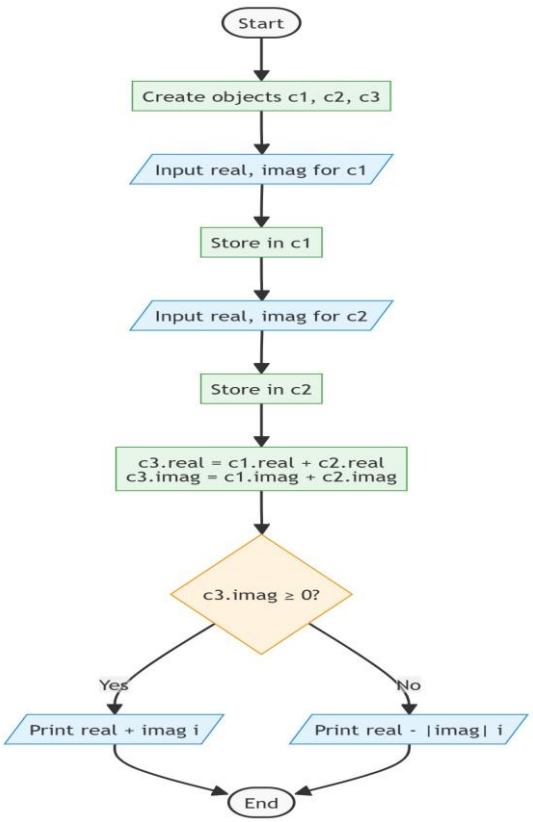
Algorithm:

1. Create class Complex

2. In Complex class:
   - initComplex(): read real and imaginary parts
   - sum(c1, c2): add real and imaginary parts separately
   - display(): print as "a + bi" or "a - bi"

3. In main:
   - Create 3 objects: c1, c2, c3
   - Input first number into c1
   - Input second number into c2
   - Add c1 + c2 and store in c3
   - Display c3

FLOWCHART:



CODE:

```
class Complex:
    def __init__(self):
        self.real = 0
        self.imag = 0
    def initComplex(self):
        self.real, self.imag = map(int, input().split())

    def sum(self, c1, c2):
        self.real = c1.real + c2.real
        self.imag = c1.imag + c2.imag
    def display(self):
        if self.imag >= 0:
            print(f"{self.real} + {self.imag}i")
        else:
            print(f"{self.real} - {abs(self.imag)}i")
c1 = Complex()
c2 = Complex()
c3 = Complex()
c1.initComplex()
c2.initComplex()
c3.sum(c1, c2)
c3.display()
```